Shivam Patel

M.S. Computer Science | Data Science and Machine Learning (Class of 2027)
patelshivamus@gmail.com | +1 (469)-826-0700 | San Diego, CA | LinkedIn | Github | Portfolio

Summary

M.S. Computer Science student (GPA **3.95/4**, SDSU) focused on NLP and applied ML. Built and deployed an AWS-hosted LLM Q&A app over PDFs; completed ML internships in sentiment analysis, healthcare text classification, and pricing prediction. Seeking full-time Data Science / ML roles starting Summer 2027.

Education

M.S. Computer Science - San Diego State University, San Diego, CA *GPA: 3.95/4 | Expected May 2027*
B.Tech Computer Science and Engineering - Indus University, Ahmedabad, India *CGPA: 9.73/10 | May 2025*

Experience

Instructional Student Assistant (ISA), Computer Networks | *San Diego State University* *Aug 2026 - Present*

- Support Computer Networks students with course concepts; grade assignments and exams.

Data Science Intern | *Embrizon Technologies* *Oct 2024 - Dec 2024*

- **Financial Sentiment Analysis:** Built a sentiment pipeline on **80K+** tweets across 10 stocks, automating model retraining.
- **Smart Healthcare Assistant:** Merged 4 datasets **(~5K records**) into a TF-IDF + Linear SVC classifier with explainability and severity scoring to reduce manual review.

Machine Learning Intern | *LogicSpark Technology* *Jun 2024 - Aug 2024*

- Integrated a **Random Forest** hotel room-pricing model using location and booking-period data for dynamic pricing.
- Automated daily margin analysis from partner booking requests.

Teaching Assistant | *Indus University* *Jan 2023 - Dec 2023*

- Supported students in ML, Data Science and Calculus with Python, data processing and problem-solving guidance.

Projects

PDF Chatbot with LLM ↗ | *Python, Flask, LangChain, FAISS, OpenAI API, Docker, AWS*

- Built and deployed a Dockerized RAG PDF chatbot on AWS via CI/CD using OpenAI embeddings, FAISS/MMR retrieval, and GPT-4o-mini; supported PDFs up to **200 MB.**
- Evaluated **85** questions across 14 PDFs, achieving **98.6%** top-5 retrieval, **85.7%** automated grounded-answer pass rate, **15/15** unsupported queries refused, and 1.33-second median latency.

NYC Collision Analysis ↗ | *Python, Pandas, NumPy, Matplotlib, Seaborn*

- Analyzed **238K+** NYC collisions: driver inattention led; March injuries ran 20% above average.
- Benchmarked three injury classifiers; logistic regression achieved **80% recall** and 0.60 F1.

Global Volcano Activity Analysis and Modeling ↗ | *Python, NumPy, Scikit-learn, XGBoost, GeoPandas, Plotly*

- Analyzed **10,000+** eruptions across **12,000 years**; engineered features for 3-class VEI classification.
- Tuned XGBoost with SMOTE, achieving **85.5%** accuracy and **0.795** macro-F1.

Employee Management System ↗ | *React, Express.js, MongoDB, Mongoose, Webpack*

- Built a full-stack CRUD application using **React, Express, and MongoDB** to manage **1,000+** employee records.

Technical Skills

Languages: Python, SQL, Java, C++, JavaScript
ML / NLP: scikit-learn, XGBoost, TF-IDF, feature engineering, model evaluation, time-series analysis, OpenCV
LLM / RAG: LangChain, FAISS, OpenAI API
Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, GeoPandas, Tableau, Streamlit
Cloud & Tools: AWS, Docker, Flask, REST APIs, CI/CD, Git, MySQL, MongoDB, Jupyter

Certifications & Honors

AWS ML Foundations | AWS ML for NLP | Google Cloud Foundations: Data/ML/AI, Infrastructure, and Data Analytics
International Math Olympiad: 3x Gold, 1x Silver, **Top 100** State Rank